

# Danium Shahnawaz Syed

Applied Cybersecurity

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## Summary

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Passionate Security Engineer and Erasmus Mundus Scholar. Solid experience in DevOps, System Administration, IIoT and privacy/security research. Holding a double Master's degree in Cybersecurity and a CCNA certification.

## Education

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**Kadir Has University & SRH Berlin University of Applied Sciences**, Erasmus Mundus Master's in Applied Cybersecurity (CyberMACS) – Istanbul & Berlin Sept 2024 – Sept 2026

- Selected among the top 32 students from over 1,000 applications
- Received a fully funded scholarship given to the top applicants
- GPA: 3.48/4.00 (KHAS), 1.8 (SRH)

**American International University-Bangladesh (AIUB)**, BSc in Electrical and Electronics Engineering – Dhaka, Bangladesh Sept 2019 – Sept 2023

- Received multiple awards and studied on a full scholarship since admission
- Graduated top of class with a CGPA of 3.98/4.00 (1.03 German scale)
- Received Best Project Award for Thesis/Capstone Project

**Mount Royal Academy**, IGCSE and A-Levels in Pearson Edexcel British Curriculum – Bangladesh June 2017 – June 2019

## Experience

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**Cybersecurity Intern**, Cyberforce – Istanbul, Turkey June 2025 – Sept 2025

- Developed internal tools for security testing of LLM/ML models from PyTorch, TensorFlow, Keras, Sklearn, and XGBoost, detecting insecure serialization formats inside Docker containers and malware-corrupted models to prevent model serialization attacks
- Conducted on-site and offline penetration testing of ML and LLM models; completed all levels of the Gandalf and Prompt Airlines CTFs
- Conducted security research on Telecom Networks and IoT networks, using the HackRF One SDR

**System Administrator**, SECOM Technologies Ltd. – Dhaka, Bangladesh Mar 2023 – Aug 2024

- Managed Linux, macOS, and Windows virtual machines supporting internal business applications
- Administered Proxmox virtualization infrastructure and used Ansible playbooks

**ICT and DevOps Intern**, Grameenphone (Telenor) – Dhaka, Bangladesh Jan 2023 – June 2023

- Automated ICT workflows and built an internal tool to monitor monthly product revenues across 8 products, presenting them on a dashboard
- Worked on the system architecture formation of Grameenphone Alo, a proprietary IoT platform
- Extensively worked on hybrid-cloud architectures and Kubernetes, and on secure CI/CD deployment through Jenkins, Docker, and Terraform
- Conducted on-site surveys of Grameenphone datacenters

**IT Educator**, Dignitus International School – Narayanganj, Bangladesh

July 2017 – July 2019

- Educated students from grade 8 to 10, with multiple A\* achievements in taught subjects for board exams
- Maintained lesson plans and liaised with the international education board Edexcel

## Publications

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### **Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains**

Nov 2026

*Danium S. Syed*

(Submitted to SECITC (Springer LNCS) conference proceedings)

### **Analyzing Political Sentiment on Twitter with Zero-Shot Large Language Models**

Oct 2026

R. Ayoub, *Danium S. Syed*, E. O. Njoku

(Accepted in Proc. IACyC 2026)

### **Explainable Botnet Detection Using Flow-Based Random Forest Classification and OSINT Validation**

Apr 2026

A. Razan, E. O. Njoku, *Danium S. Syed*, N. Nachevski

(Web Proc. CIIT 2026)

### **Reputation and Energy-Aware Dynamic Hybrid Consensus (READ-HC) Model for IIoT**

July 2025

S. Mitra, V. S. Sable, *Danium S. Syed*

[10.31585/jbba-8-2-\(3\)2025](https://doi.org/10.31585/jbba-8-2-(3)2025) (The Journal of The British Blockchain Association)

### **Design and Feasibility Analysis of a LoRa Based Communication System for Disaster Management**

Aug 2024

*Danium S. Syed*, S. B. Hossain, A. Islam, T. A. I. Jim, M. Saniat, R. Zishan

[10.53799/ajse.v23i2.1055](https://doi.org/10.53799/ajse.v23i2.1055) (AIUB Journal of Science and Engineering (AJSE), 23, 135-144)

## Projects

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### **Master's Thesis: Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains**

Jan 2026 – present

- Conducted security analysis of containerized ML systems to measure prevalence and distribution of embedded model artifacts across Docker ecosystems
- Identified security risks in ML supply chains by statically analyzing model files (PyTorch, TensorFlow, ONNX, Pickle) for unsafe serialization and code execution vulnerabilities
- Built an automated pipeline to extract container filesystems and scan ML artifacts, generating empirical insights into real-world deployment risks

### **Linux Home Server (HomeLab)**

March 2020 - Ongoing

- Built from scratch and maintains a Debian-based home server for security research, home automation, and media consumption
- Follows DevOps practices and maintains high uptime
- Technology stack: WireGuard, Docker Compose, Proxmox

### **Bachelor's Capstone: Long-Range Off-Grid Communication System Deployed with Amphibious Rovers**

Jan 2023 – Sept 2023

- Built a mesh communication system using LoRa modulation for deployment in disaster scenarios via amphibious rovers
- Provides text messaging and location-based communication services for affected users in disaster zones

More projects: [daniumsyed.github.io/projects](https://daniumsyed.github.io/projects)

## Honors and Awards

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- 5th Place (All Solves), CYBERUS Embedded Systems CTF Competition — Lorient, France (2026)
- 2nd Place (All Solves), CyberMACS Winter School CTF Competition — Istanbul, Turkey (2025)
- Erasmus Mundus Full Scholarship, EMJM in Applied Cybersecurity — Turkey/Germany (2024)
- Dean's Best Project Award, AIUB — Dhaka, Bangladesh (2023)
- Dean's List Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Freedom Fighter's Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Edexcel High Achiever's Award, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)
- Featured in The Daily Star and The Daily Sun, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)

## Certifications

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- Building Resilience: Navigating Vulnerability, Risks, and Continuity in a Dynamic World — CyberMACS (2026)
- Cisco Certified Network Associate (CCNA), CSC014934967 — Cisco (2025)
- The Bits and Bytes of Computer Networking — Google/Cisco, Coursera (2020)

## Skills

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**DevOps:** AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

**System Administration:** Linux, Networking, Virtualization, Containers, Git

**Privacy & AI Security:** Differential Privacy, Agentic AI, PyTorch, TensorFlow, vLLM/Ollama

**Security Research:** Vulnerability Analysis, Threat Modeling (MITRE ATT&CK), Penetration Testing (Hardware and Software), Incident Response

**Scripting:** Python, JavaScript, Bash, LaTeX

**Languages:** English (IELTS 8.0), Bengali (Native), Turkish (A1), German (A1, ongoing studies)